

Backtest Discipline: Three Traps That Inflate Performance

Look-ahead, survivorship, multiple-testing, measured on synthetic series (not investment advice)

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Abstract

A backtest that looks great and then falls apart in live trading is usually not evidence that the market changed regime. Far more often it is evidence that the backtest itself contains a structural flaw that inflates apparent performance. This paper isolates three of the most commonly cited inflation mechanisms, look-ahead bias, survivorship bias, and multiple-testing-driven data snooping, in controlled experiments and measures how large the resulting illusion is. Every experiment runs on synthetic random-walk price series (geometric Brownian motion, GBM) generated locally; no real market data was used, and the goal is not to claim any trading strategy is profitable but to make the inflation mechanisms themselves visible as numbers. A single same-bar look-ahead bug alone pushed the annualized Sharpe ratio to roughly 21.1; lagging the signal by one day, the honest version, brought it down to 0.07. A full universe that honestly includes random delistings produced a mean five-year CAGR of 10.1%, while a survivors-only backtest built from today's constituent list on the same data produced 13.9%, a 3.8 percentage point inflation. Scanning more random moving-average-crossover parameter combinations on the same single series, from 5 up to 2,000, smoothly raised the best in-sample Sharpe ratio from 0.34 to 0.90, while the exact same chosen strategy, evaluated strictly out-of-sample, stayed near a noise floor of zero (-0.07 to 0.17) across every trial count. These results are a methodology demonstration and not investment advice. No claim is made about the profitability of any real trading strategy, and every number in this paper comes from synthetic data.

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1 Abstract

A backtest that looks great and then falls apart in live trading is usually not evidence that the market changed regime. Far more often it is evidence that the backtest itself contains a structural flaw that inflates apparent performance. This paper isolates three of the most commonly cited inflation mechanisms, look-ahead bias, survivorship bias, and multiple-testing-driven data snooping, in controlled experiments and measures how large the resulting illusion is. Every experiment runs on synthetic random-walk price series (geometric Brownian motion, GBM) generated locally; no real market data was used, and the goal is not to claim any trading strategy is profitable but to make the inflation mechanisms themselves visible as numbers. A single same-bar look-ahead bug alone pushed the annualized Sharpe ratio to roughly 21.1; lagging the signal by one day, the honest version, brought it down to 0.07. A full universe that honestly includes random delistings produced a mean five-year CAGR of 10.1%, while a survivors-only backtest built from today's constituent list on the same data produced 13.9%, a 3.8 percentage point inflation. Scanning more random moving-average-crossover parameter combinations on the same single series, from 5 up to 2,000, smoothly raised the best in-sample Sharpe ratio from 0.34 to 0.90, while the exact same chosen strategy, evaluated strictly out-of-sample, stayed near a noise floor of zero (-0.07 to 0.17) across every trial count. These results are a methodology demonstration and not investment advice. No claim is made about the profitability of any real trading strategy, and every number in this paper comes from synthetic data.

2 Background: Backtests Fail in Practice Because of Method, Not Markets

Anyone who has built a quantitative trading strategy has seen this scene. A backtest window shows an elegant equity curve with a Sharpe ratio above 2. The strategy goes live, a few months pass, and the curve is nowhere to be found. The most common explanation offered is “the market regime changed.” Regimes genuinely do shift. But a substantial share of the performance collapse observed in practice comes from something far more mundane and far more fixable: a structural flaw baked into how the backtest itself was constructed.

The distinction matters because the remedy is completely different. If a regime shift caused the collapse, the answer is a harder problem: a more robust model, more data, better risk management. If a methodological flaw caused it, the answer is comparatively simple: confirm that every signal only uses information that was actually knowable at that point in time, recompute using the full universe that existed back then rather than today's survivors, and count how many parameter combinations were tried so that count can be statistically corrected for. The difficulty is that these flaws mostly hide quietly inside backtest code. The code runs without error, the numbers look plausible, and nothing warns the researcher that a flaw is present.

This paper picks three of the flaws most frequently cited in both academic and practitioner literature, look-ahead bias, survivorship bias, and multiple testing, and isolates each in its own controlled experiment. Mixing all three into one experiment would make it impossible to tell how much each contributed, so each is deliberately demonstrated on separate synthetic data that does not interact with the others. None of the three experiments uses real tickers or real market data. Each uses a pure random walk generated with numpy, whose true expected return and true volatility we set ourselves. Knowing the “right answer” in advance is exactly why synthetic data is useful here: with real market data there is no way to know the ground truth, so there is no way to isolate how much a given backtesting method distorts it. With data whose generating process we control, that isolation becomes possible.

3 Three Inflation Mechanisms

3.1 Look-Ahead Bias: Code That Peeks at the Future

Look-ahead bias covers every case where a backtest's signal calculation uses information that did not yet exist at that point in time. The most blatant form is an indexing bug that accidentally uses the same bar's closing price for both the signal and the return it is scored against. Implement “go long today if today's close is above yesterday's, short if below” as a vectorized rule without care, and the information deciding the signal (today's close) arrives at the exact same moment as the return the signal is scored on (also measured from today's close).

The rule is effectively “know whether today will go up or down and enter in that direction before it happens”, an access to information that is impossible in live trading but appears naturally the moment an array index is off by one. Subtler forms include using a financial statement’s fiscal period-end date instead of its actual public release date, using vendor data that has since been revised as if it were the originally observed value, or assuming fills at the exact closing price with no slippage. What all of these share is that information the market did not yet have leaks into the backtest.

3.2 Survivorship Bias: Dead Tickers Quietly Disappear from the Data

Survivorship bias is a problem with how the universe itself is defined: it is built with the benefit of hindsight already baked in. Most market data vendors provide a list of tickers that are still trading today. Names that were delisted, acquired, or went bankrupt are quietly absent from that “current constituents” list. When an analyst takes that current list and runs a backtest five years into the past, the hundreds of names that existed during that period but no longer exist today are simply missing from the calculation. This is not a random omission. The names that disappeared are systematically concentrated among the worst performers, because bankruptcy and delisting do not happen to companies that are doing well. The result is that a survivors-only average return is always biased upward relative to what an investor who actually held that market at the time would have earned. This mechanism has long been documented in mutual fund performance research (Brown, Goetzmann, Ibbotson & Ross, 1992), and the same mechanism operates identically in single-stock backtests.

3.3 Multiple Testing: Try Enough Things and Something Will Look Good by Chance

Multiple testing, commonly called data snooping, is different in kind from the first two. Look-ahead bias and survivorship bias are coding mistakes; multiple testing happens even when every line of code is correct. The failure is not counting how many things were tried. Scan only five moving-average-crossover parameter combinations on a single price series and one might happen to look good in-sample by chance; scan 2,000 and something spectacular-looking is almost guaranteed to appear. This is not because the series actually contains that pattern, but purely because the parameter search picked out whichever combination happened to be luckiest within the sample’s own noise. If 2,000 people each flip a coin 100 times, a handful of them will see an astonishing run of 15 heads in a row. That person has no special coin-flipping skill; the extreme outcome is guaranteed to appear somewhere once enough people try. Parameter scanning in a backtest has the exact same structure. As the number of trials N grows, the expected value of “the best in-sample result” keeps growing even when there is zero true effect anywhere in the data. Reporting only that single best result as a “discovered strategy” without correcting for how many trials produced it is exactly what inflates performance through multiple testing (Bailey, Borwein, Lopez de Prado & Zhu, 2014).

4 Experimental Setup: Isolating Each Mechanism on Synthetic Series

All three experiments run on synthetic price series generated locally with numpy alone; none uses real market data or any network call. Each experiment repeats across 30 independent random seeds, and both the mean and standard deviation are reported.

Look-ahead experiment. A 1,000-trading-day price path is drawn from a geometric Brownian motion with 6% annualized expected return and 20% annualized volatility. The buggy strategy uses that day’s own return sign as that day’s position ($\text{position} = \text{sign}(\text{today's return})$, $\text{pnl} = \text{position times today's return}$), which requires knowing today’s close before it happens; multiplying a return by its own sign always yields the absolute value of that return, so losses are structurally impossible. The fixed strategy uses yesterday’s return sign as today’s position ($\text{position} = \text{sign}(\text{yesterday's return})$). This is an executable backtest, but a pure random walk has no autocorrelation, so no genuine edge is expected from it.

Survivorship experiment. 500 independent stocks are simulated over five years (1,260 trading days). Each stock’s annualized expected return is drawn from a distribution with mean 6% and cross-sectional standard deviation 12%; annualized volatility is fixed at 28% for all. Any stock whose drawdown from its own running

peak exceeds 60% faces a 1% daily probability of delisting, and once delisted its price is frozen permanently at that level (an unrecoverable loss). The full-universe calculation holds all 500 stocks equal-weighted from day one through the final day. The survivors-only calculation keeps only the subset still listed at the end and rebuilds an equal-weighted portfolio from just that subset, mimicking the common practice of backtesting with a data vendor that only supplies today’s constituent list.

Multiple-testing experiment. A zero-drift (0% annualized expected return, 20% annualized volatility) random walk of 1,200 trading days is split into an in-sample segment (the first 60%, 720 days) and an out-of-sample segment (the remaining 40%, 480 days). N random (fast moving average, slow moving average) parameter pairs are drawn ($N = 5, 20, 100, 500, 2,000$), each pair’s in-sample Sharpe ratio is computed, and the single best-performing pair is selected. That exact same parameter pair is then applied unchanged to the out-of-sample segment to measure how it actually performs. The true expected return was fixed at exactly zero by design so that the ground truth is known in advance; the experiment measures how convincing an in-sample result can be “discovered” in a series with no real edge whatsoever. Two corrections are applied jointly. The first is a Bonferroni-style correction that divides the single-test significance level of 0.05 by the number of trials N , a conservative adjustment. The second is the deflated Sharpe ratio (DSR) proposed by Bailey and Lopez de Prado (2014), which uses an extreme-value approximation to estimate the “best Sharpe ratio” expected under the null across N independent trials, then converts how far the observed best Sharpe ratio exceeds that baseline into a standard normal probability.

5 Results

5.1 Look-Ahead Bias: A Single Index Shift Produces a 21x Illusion

The buggy strategy using same-bar information averaged an annualized Sharpe ratio of 21.1 (standard deviation 0.59) across the 30 seeds. No real strategy achieves anything close to this; the value exists only because the design makes losses structurally impossible. Lagging the signal by one honest day drops the mean Sharpe ratio to 0.069 (standard deviation 0.49), scattered noise around zero, exactly consistent with the fact that a random walk has no genuine autocorrelation to exploit. Comparing the two in CAGR terms is even more dramatic: the buggy strategy’s mean CAGR was 1,132% (an 11.3x annual multiple), and the fixed strategy’s mean CAGR was -0.15%, essentially exactly zero.

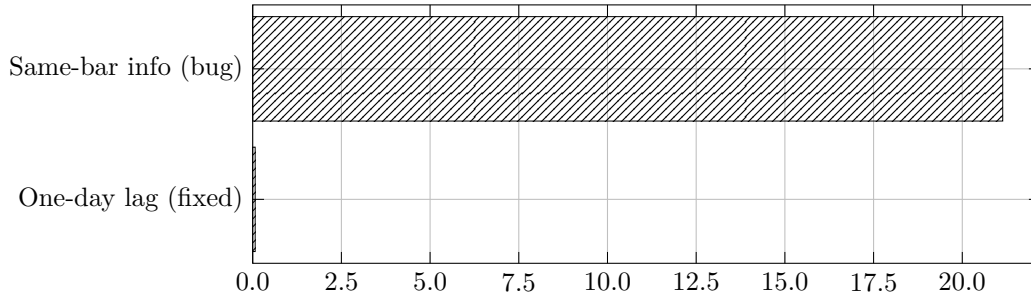


Figure 1: Annualized Sharpe: same-bar bug vs one-day lag

The implication is simple: look-ahead bias is not a matter of degree, it is a matter of existence. Once same-time information contaminates both the signal and the return it is scored against, the illusion can be a 21x inflation or a 1.3x inflation depending on how much contamination occurs. Real-world look-ahead bugs are rarely this blatant; they typically shift the index by hours rather than a full day, or by the length of a single delayed filing. The subtler the form, the harder it is to catch, and until it is caught it produces a backtest result that reads as entirely honest.

5.2 Survivorship Bias: Counting Only Survivors Erases 3.8 Percentage Points

Across the 500-stock simulation, an average of 101.5 stocks (about 20.3%) were delisted over the five years. The full-universe mean CAGR, honestly including the realized losses of delisted names, was 10.1% (standard

deviation 0.94 percentage points). The survivors-only mean CAGR, recomputed using only the 398.5 names still listed at the end, was 13.9% (standard deviation 0.98 percentage points). The gap between the two, the pure illusion created by survivorship bias, is 3.8 percentage points per year.

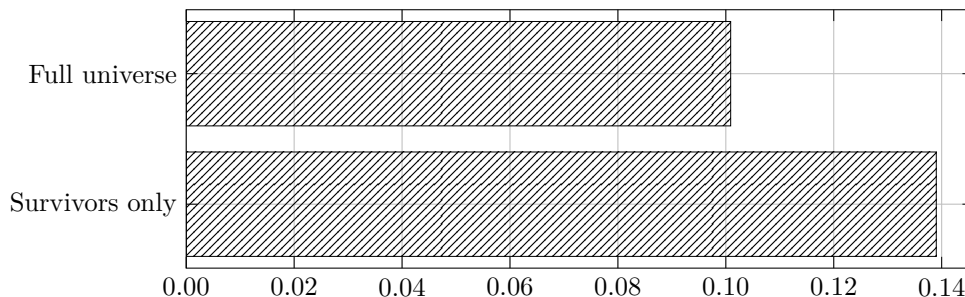


Figure 2: Five-year CAGR: full universe vs survivors-only

3.8 percentage points may look small, but it compounds annually over multiple years. Assuming a 20-year holding period, the gap between a 10.1% and a 13.9% CAGR compounds into roughly double the final portfolio value. What matters more is how the delisting probability was designed: it was tied directly to drawdown, so poorly performing stocks were more likely to be delisted. This is exactly the direction delisting, acquisition, and bankruptcy occur in real markets. Thriving companies rarely get delisted suddenly; companies whose fundamentals have collapsed are delisted overwhelmingly more often. So filtering to survivors is not a random omission but a systematic removal of the worst observations, and that directionality is what determines the size of the bias.

5.3 Multiple Testing: In-Sample Performance Keeps Climbing, True Performance Does Not

The multiple-testing results are summarized below. The best in-sample Sharpe ratio rises smoothly as the number of trials N grows, while the exact same chosen strategy’s out-of-sample Sharpe ratio stays near zero across all five trial counts.

Trials N	Best in-sample Sharpe (ann.)	Matched out-of-sample Sharpe (ann.)	Expected best Sharpe under null	Naive p-value	Bonferroni rejection rate (5%)	DSR “no skill” probability
5	0.34	0.17	0.71	0.316	0.0%	0.694
20	0.61	0.09	1.12	0.196	0.0%	0.767
100	0.78	-0.04	1.50	0.138	0.0%	0.840
500	0.85	0.01	1.81	0.114	0.0%	0.911
2,000	0.90	-0.07	2.04	0.103	0.0%	0.946

Three things need to be read together for this table to make sense. First, the best in-sample Sharpe ratio rises from 0.34 at $N=5$ to 0.90 at $N=2,000$, nearly a threefold increase. Because the true expected return of this series was fixed at exactly zero by design, this increase reflects no improvement in any strategy; it is purely the effect of searching harder and having more chances to stumble onto something that looks good by luck. Second, the exact same parameter pair, evaluated out-of-sample, stays within a noise band of -0.07 to 0.17 regardless of N . Out-of-sample performance not improving alongside N means the “edge” found in-sample transfers to the out-of-sample period not at all. Third, a naive significance test that ignores how many trials were run (a single-test 5% threshold) calls the result “statistically significant” increasingly often as N grows, from 6.7% of seeds up to 46.7%, exactly the failure pattern multiple-testing theory predicts. The Bonferroni correction (dividing the significance threshold by N) and the deflated Sharpe ratio, by contrast, hold their rejection rate at 0% across every value of N , correctly identifying that this data was constructed with zero true edge. The DSR’s “no skill” probability also rises from 0.69 to 0.95 as N grows, exactly quantifying the intuition that an observed best result should be treated more skeptically the more trials produced it.

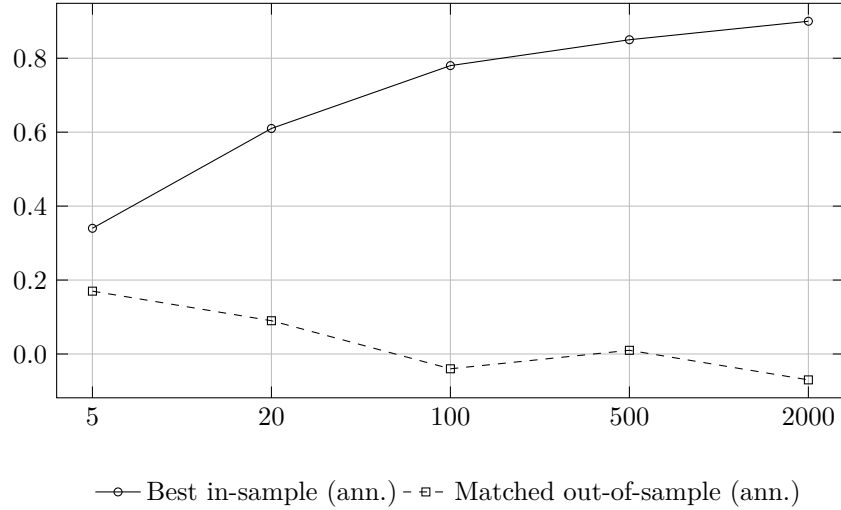


Figure 3: Best in-sample vs matched out-of-sample Sharpe by trial count N

6 A Practical Checklist: The Minimum Procedure to Guard Against All Three

What all three experiments point toward is not a sophisticated new technique but a boring procedure. Checking the following items when reviewing a backtest catches most instances of all three inflation mechanisms.

- **Guarding against look-ahead bias.** Explicitly verify that the index used to compute a signal differs from the index used to compute the return it is scored against. Use a financial statement’s actual public release date, never its fiscal period-end date. If vendor data has since been revised, keep a separate point-in-time copy of what was actually observable at that moment rather than the later-revised value. Building a hard-coded one-bar lag layer between signal computation and return computation into the backtesting framework removes the possibility of a human accidentally shifting an index.
- **Guarding against survivorship bias.** Use a point-in-time universe for each date rather than a vendor’s “currently listed” list. Always include the last traded price or acquisition price of delisted or acquired names in the final return calculation. Confirm at the vendor contract stage whether the provided universe is survivorship-bias-free; if the documentation says “current constituents only,” assume any backtest built on that data is structurally biased.
- **Guarding against multiple testing.** Track the total number of parameter combinations, variants, and features tried from the very start of the research process. Re-evaluate the final reported Sharpe ratio against a trial-count-adjusted threshold (Bonferroni or deflated Sharpe ratio) rather than a single-test threshold. Confirm that a strategy selected in-sample is reproduced on a truly separate out-of-sample segment, ideally on data observed only after the split was fixed. Remember that repeatedly cycling between in-sample tuning and out-of-sample checking on the same data (“it didn’t work out-of-sample, so let’s go back and retune”) effectively contaminates the out-of-sample segment with multiple testing as well.

7 Reproducibility

All three experiments run on numpy and scipy alone, with no external network calls and no real market data of any kind. Each was repeated across 30 independent random seeds, with both the mean and standard deviation reported, and every metric (Sharpe ratio, CAGR, in-sample and out-of-sample performance, p-value, deflated Sharpe ratio) was computed directly by code, never self-reported by a model or agent. All three experiments together finish in a few seconds on a laptop CPU. The deflated Sharpe ratio implements the approximation from Bailey and Lopez de Prado (2014) directly, and the expected best Sharpe ratio under the null uses the extreme-value approximation that includes the Euler-Mascheroni constant. The seeds, parameters, and synthetic data

generation process needed to reproduce every number are all fixed in this paper’s experiment script.

Every number in this paper comes from a methodology demonstration on synthetic random-walk data, and no claim is made about the profitability of any real trading strategy or any market forecast. **This content is research and methodology only, and is not investment advice.**

8 References

1. Bailey, D. H., & Lopez de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management*, 40(5).
2. Bailey, D. H., Borwein, J., Lopez de Prado, M., & Zhu, Q. J. (2014). Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance. *Notices of the American Mathematical Society*, 61(5).
3. Brown, S. J., Goetzmann, W., Ibbotson, R. G., & Ross, S. A. (1992). Survivorship Bias in Performance Studies. *Review of Financial Studies*, 5(4).
4. Harvey, C. R., Liu, Y., & Zhu, H. (2016). ...and the Cross-Section of Expected Returns. *Review of Financial Studies*, 29(1).
5. Aronson, D. (2006). *Evidence-Based Technical Analysis: Applying the Scientific Method and Statistical Inference to Trading Signals*. Wiley.
6. Pardo, R. (2008). *The Evaluation and Optimization of Trading Strategies*, 2nd ed. Wiley.